

The Coordinate Plane

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

CUE FOCUS The plane has four quadrants formed by the x-axis and y-axis, meeting at the origin (0, 0). Plot an ordered pair (x, y) by going over on x first, then up or down on y. Sign patterns: I (+, +), II (–, +), III (–, –), IV (+, –).

Round 1 — Retrieve It

1. How many quadrants does the coordinate plane have, and what two axes form them?

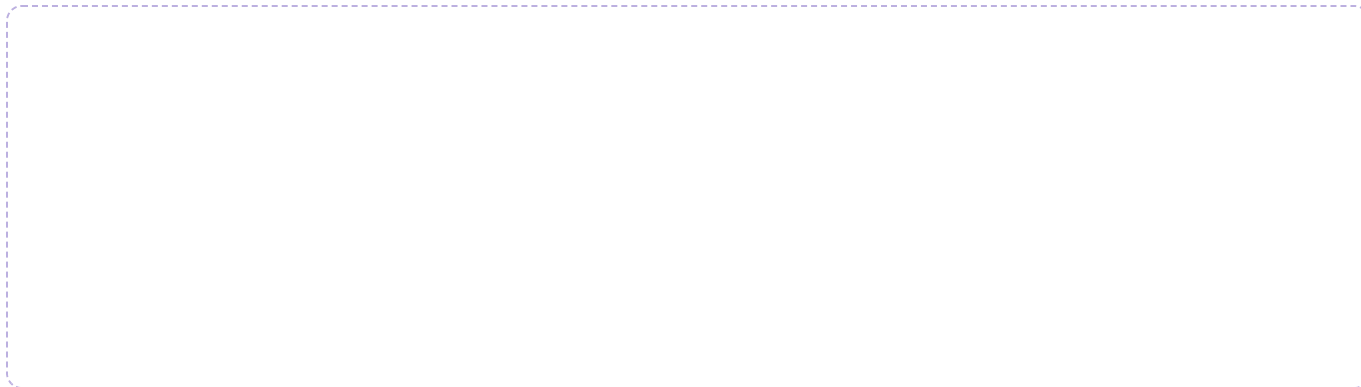
2. What are the coordinates of the origin?

3. To plot (x, y), which coordinate do you move on first?

4. Give the sign pattern (x, y) for each quadrant I–IV.

Round 2 — Sketch It

Draw and label the x-axis, y-axis, and origin. Number the four quadrants I–IV, then plot and label the point (3, –2). Circle the quadrant it lands in.



The Coordinate Plane — Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 — Apply the Relationship

1. In which quadrant is the point $(-4, 5)$?

2. Describe the steps to plot $(-3, -6)$ from the origin.

3. A point is 5 units right and 2 units up from the origin. Write its ordered pair.

Round 4 — Reason from Evidence

A student plotted $(2, -5)$ in Quadrant I. Use the sign patterns to explain the mistake, name the correct quadrant, and tell how the signs reveal it.

The Coordinate Plane — ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain why order matters in an ordered pair, using (3, 5) and (5, 3) as examples not printed on the cue card.

C**Connect**

Connect the coordinate plane to a map or a video game grid. How are x and y like directions such as "right" and "up"?

E**Extend**

Create four ordered pairs, one in each quadrant. List each point with its quadrant and explain how the signs place it there.

Confidence check. Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it.

Teacher Key — The Coordinate Plane

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. Four quadrants, formed by the x-axis and the y-axis.
2. **(0, 0)**.
3. Move on the x-coordinate first (left or right), then y (up or down).
4. I (+, +), II (-, +), III (-, -), IV (+, -).

Sketch It — look for

Axes and origin labeled, quadrants numbered I–IV counterclockwise from top right. Point (3, -2) is right 3, down 2, landing in **Quadrant IV**.

Apply It

1. (-4, 5): negative x, positive y → **Quadrant II**.
2. From the origin, move left 3 ($x = -3$), then down 6 ($y = -6$); lands in Quadrant III.
3. Right 5, up 2 → **(5, 2)**.

Reason from Evidence — look for

(2, -5) has a positive x and a negative y, which is the pattern (+, -) for **Quadrant IV**, not Quadrant I. Quadrant I needs both coordinates positive.

ACE — Extend look-for

Many answers possible. Check that one point falls in each quadrant and that its signs match the quadrant pattern.

Suggested use. Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.